

ABSTRACT

Maintaining hygiene and safety in public restrooms is a significant challenge, especially in high-traffic areas where frequent human contact increases the risk of contamination. This paper introduces a hygiene-ensured smart door locking system designed to automate restroom operations using sensor-based control and microcontroller technology. The system employs an Arduino as the central controller, interfaced with infrared (IR) sensors, a servo motor, an emergency alert module, and a digital display to create a fully automated and touch-free restroom experience. When a user approaches the restroom, the entrance IR sensor detects their presence and triggers the servo motor to open the door automatically. Once the person enters, the door closes to ensure privacy and remains locked until the flushing process is completed. A second IR sensor detects toilet usage and activates an automatic flush cycle, ensuring cleanliness after every use. The door only reopens once the flushing process finishes, maintaining both hygiene and operational safety. An emergency switch and buzzer system are incorporated to alert nearby personnel in case of distress, enhancing user safety. Additionally, a 16×2 LCD displays real-time information, including door status, flush operation, and emergency alerts. By minimizing physical contact, the proposed system significantly reduces the spread of germs, ensures proper sequencing of restroom functions, and promotes a safer and more hygienic environment. The proposed design is particularly suitable for smart infrastructure applications such as hospitals, schools, and public facilities, where hygiene and automation are of utmost importance.

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CHAPTER 1

INTRODUCTION

Maintaining hygiene and safety in public and shared restrooms has become a growing concern in modern society, especially in the context of increasing health awareness and technological advancement. Public toilets are among the most frequently used facilities in urban environments, yet they are also one of the primary sources of cross-contamination due to constant human interaction with surfaces such as doors, flush handles, and taps. The transmission of pathogens through these contact points contributes significantly to the spread of infectious diseases, posing serious health risks in community and healthcare settings. As a result, the development of automated, touchless, and intelligent systems for restroom management has gained considerable attention in recent years. The proposed hygiene-ensured smart door locking system aims to address these challenges by integrating automation, sensing, and control technologies to enhance hygiene, ensure privacy, and provide a safer restroom experience.

Conventional restroom systems rely heavily on manual operations, where users are required to physically open and close doors, operate flushing mechanisms, and manage hygiene processes independently. This manual dependency not only reduces operational efficiency but also increases the likelihood of germ transmission. In contrast, smart restroom automation introduces a paradigm shift by enabling sensor-driven interactions that minimize physical contact. The key principle behind such systems is the use of sensors and microcontrollers to detect human presence and control actuators accordingly. By leveraging simple yet effective electronic components, it becomes possible to design a cost-efficient and scalable solution that can be deployed in various public and private infrastructures.

The hygiene-ensured smart door locking system presented in this study utilizes an Arduino microcontroller as the central control unit. Arduino is widely known for its flexibility, ease of programming, and compatibility with different sensors and actuators. In the proposed system, the Arduino is interfaced with two infrared (IR) sensors, a servo motor, an emergency switch, a buzzer, and a 16×2 Liquid Crystal Display (LCD). Each component plays a vital role in the functionality of the system. The first IR sensor,

positioned at the restroom entrance, detects when a person approaches the door. Upon detection, the Arduino triggers the servo motor to open the door automatically, allowing the user to enter without touching any surface. After a predefined delay, the door closes to ensure privacy and restricts access to others while the restroom is occupied.

Once the user is inside, the second IR sensor, located near the toilet, monitors usage activity. After detecting that the user has finished, it triggers the automatic flushing mechanism, ensuring that hygiene is maintained consistently. The door remains locked during this process and only reopens once the flush cycle is complete. This sequence of events ensures that the restroom remains clean and ready for the next user while preventing accidental reentry during cleaning operations. Additionally, the system incorporates an emergency switch to enhance user safety. In the event of distress or malfunction, pressing the emergency button activates a buzzer, alerting nearby maintenance staff or security personnel to take immediate action.

The LCD display provides real-time status updates, including the door's position (open or closed), flush status (active or idle), person detection, and emergency notifications. This feature ensures transparency in operation and helps maintenance staff monitor restroom activity without invading privacy. The entire setup is compact, energy-efficient, and easy to maintain, making it a practical solution for public infrastructure such as airports, shopping complexes, hospitals, educational institutions, and smart city environments. Furthermore, the modular nature of the system allows for scalability and integration with Internet of Things (IoT) platforms in future implementations, enabling remote monitoring and predictive maintenance.

The development of this system aligns with the growing demand for automation and smart infrastructure in the context of the Fourth Industrial Revolution (Industry 4.0). Smart restrooms contribute to the broader ecosystem of intelligent buildings, where environmental control, user safety, and energy management are governed through interconnected sensors and real-time data processing. Hygiene automation is not only a matter of convenience but also a necessity in environments where public health and sanitation are critical. For instance, in hospitals and healthcare facilities, contactless restroom systems can significantly reduce the risk of nosocomial infections. Similarly, in

educational institutions and workplaces, such systems can promote better hygiene habits and reduce absenteeism caused by infectious diseases.

In terms of design philosophy, the proposed smart door locking system emphasizes simplicity, reliability, and user-centric functionality. Instead of relying on complex or expensive components, the system employs widely available electronic modules that are affordable and easy to replace. The use of Arduino ensures that the control logic can be customized or upgraded as needed. For example, the system could later incorporate proximity sensors, RFID authentication, or voice-controlled assistance to enhance accessibility for differently abled individuals. Power efficiency is another critical consideration, as the components consume minimal energy during standby and active operation, making the system suitable for continuous use in public environments.

The hygienic benefits of automation extend beyond disease prevention. By eliminating the need for physical contact with surfaces, automated restrooms also improve user comfort and public perception of cleanliness. Moreover, the automatic flushing process ensures consistent water usage and reduces wastage by controlling the duration and timing of each flush precisely. This feature contributes to environmental sustainability, aligning the project with global efforts toward eco-friendly smart city development. The emergency alert mechanism further enhances the reliability and safety of the system, ensuring that users are not trapped or endangered during unexpected technical failures.

The hygiene-ensured smart door locking system is a step toward creating intelligent, user-friendly, and hygienic restroom facilities. It integrates core elements of automation—sensing, control, and actuation to create a seamless and contactless user experience. The system's logic ensures synchronized operation between the door and flush mechanisms while providing safety through emergency alerts and transparency through digital display. The design addresses key challenges associated with manual restroom use, such as hygiene maintenance, privacy assurance, and operational efficiency.

By promoting hands-free interaction and structured automation, the system contributes to public health protection, technological advancement, and improved facility management. With further development and integration into IoT frameworks, such systems can form part of large-scale smart infrastructure networks that monitor hygiene, occupancy,

and maintenance requirements autonomously. Thus, this project not only offers an immediate solution for improving restroom hygiene but also establishes a foundation for future innovation in intelligent sanitation technologies.

1.1 OVERVIEW

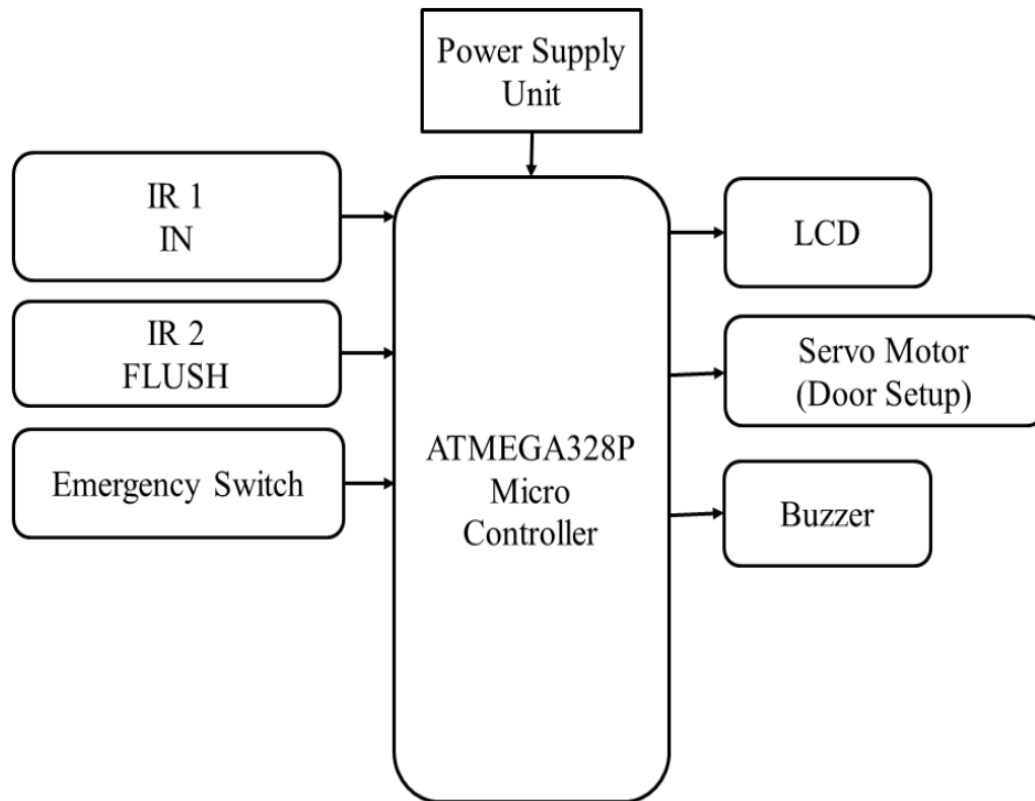


Fig 1: BLOCK DIAGRAM

The hygiene-ensured smart door locking system represents a forward-thinking solution to one of the most overlooked yet significant public health concerns—maintaining cleanliness and operational safety within restrooms. The system’s design integrates automation, sensor technology, and microcontroller-based control to achieve a fully autonomous restroom experience. This overview elaborates on the conceptual framework, functional operation, hardware integration, and advantages of the proposed system, highlighting how it contributes to hygiene management, user safety, and environmental sustainability. By automating the restroom’s entry, flushing, and alert mechanisms, the system eliminates the need for manual handling of commonly touched surfaces, significantly minimizing the risk of contamination. The result is a more efficient, safe, and

intelligent public sanitation environment suitable for deployment in hospitals, schools, offices, and other shared spaces.

At the core of this design is the Arduino microcontroller, which serves as the system's central processing unit. Arduino is known for its flexibility, reliability, and adaptability to various sensor and actuator configurations. In this setup, it reads inputs from infrared (IR) sensors, processes the received signals according to predefined logic, and then controls the output devices, including the servo motor, LCD display, and buzzer. The system's workflow follows a logical sequence that ensures proper coordination among all components, preventing overlap or malfunction. This intelligent control loop allows seamless automation—once the user approaches the restroom, uses the facility, and exits—without requiring direct human intervention at any stage.

The infrared sensors play a crucial role in user detection and process automation. The first IR sensor, placed near the restroom entrance, detects an approaching individual and signals the Arduino to activate the servo motor responsible for door movement. The motor rotates the door from its closed position (0°) to its open position (90°), allowing entry. After a short delay, the door automatically closes, providing privacy for the occupant. The second IR sensor, installed near the flushing unit, monitors usage activity. When it detects that the user has finished, it initiates the automatic flushing sequence, ensuring that cleanliness is maintained consistently. The door remains closed throughout this phase, reopening only after the flushing process is complete, thereby avoiding premature access and maintaining hygienic conditions.

A servo motor is selected for door operation because of its precision and controllable angular motion. Servo motors are reliable, energy-efficient, and capable of operating smoothly within defined limits, making them ideal for controlled movement applications. By programming the servo motor to operate between 0° and 90° , the system achieves accurate door positioning, ensuring complete closure and smooth opening. This mechanism not only enhances the convenience for users but also prevents contamination caused by touching door handles or push buttons.

The emergency alert system adds a critical safety dimension to the design. Public restrooms, particularly in hospitals and old-age homes, may host users who require

immediate assistance in emergencies such as fainting, health distress, or accidental locking. To address this, an emergency push button is integrated into the circuit. When pressed, it activates a buzzer, generating an audible alert to notify nearby personnel or maintenance staff. This feature ensures user safety by enabling prompt assistance, which is essential in sensitive environments such as healthcare facilities.

A 16×2 Liquid Crystal Display (LCD) is incorporated into the system to display real-time operational data, including door status, flush activity, person detection, and emergency notifications. The LCD enhances system transparency by providing instant feedback to users and maintenance staff. For instance, it can display messages such as “Door Open,” “Door Locked,” “Flushing in Progress,” or “Emergency Alert Activated.” This feedback not only improves user experience but also allows easy diagnosis of system status without manual inspection.

The overall working principle of the system is straightforward yet highly effective. It begins with person detection at the entrance, followed by automated door opening and closing. During occupancy, the system ensures privacy by locking the door. After usage, the flushing process is automatically activated based on IR sensor detection, followed by door unlocking once the cleaning cycle is completed. This sequence ensures that hygiene and privacy are maintained at all times. The Arduino executes the entire process through pre-programmed logic, interpreting sensor signals and issuing control commands to actuators accordingly. Each action is precisely timed to ensure synchronization between different modules.

The system’s hardware design emphasizes simplicity, modularity, and durability. All components, including sensors, servo motor, switch, buzzer, and display, are powered through the Arduino’s regulated supply, minimizing wiring complexity and energy consumption. The setup can be easily extended to include additional features such as temperature sensors, automatic air fresheners, or IoT-based monitoring modules for remote supervision. The use of low-cost and easily available components makes the system highly scalable, enabling deployment across multiple restrooms without significant cost increase.

In addition to its technical advantages, the hygiene-ensured smart door locking system provides multiple social and environmental benefits. From a health perspective, it

minimizes the transmission of germs and pathogens by reducing physical contact. This feature is particularly valuable during disease outbreaks or pandemics when limiting surface contact is vital to prevent infection spread. From an environmental standpoint, the automated flushing system ensures optimal water usage by controlling the duration and frequency of each flush. This prevents unnecessary wastage, promoting resource efficiency and sustainability. Moreover, the system's ability to operate on low power enhances its environmental friendliness and allows potential integration with solar energy systems in smart building frameworks.

The application scope of this system extends beyond public restrooms. It can be customized for use in hospitals, airports, schools, shopping complexes, offices, and even residential buildings where hygiene automation is desirable. In hospitals, it helps reduce hospital-acquired infections by preventing patients or visitors from touching contaminated surfaces. In airports and malls, it streamlines user flow and improves sanitation management. In educational institutions, it promotes hygiene awareness among students and staff, contributing to a healthier environment. Furthermore, the same concept can be integrated into smart city infrastructure, where interconnected restroom units communicate with a central monitoring system for maintenance scheduling, occupancy analytics, and fault detection.

Another significant aspect of this overview is the system's future enhancement potential. The current design operates as a standalone system controlled locally by the Arduino. However, future developments could include Internet of Things (IoT) integration for remote control, data collection, and predictive maintenance. For instance, IoT connectivity could allow facility managers to monitor restroom usage, water consumption, and emergency alerts in real time from a centralized dashboard. Machine learning algorithms could analyze this data to predict maintenance needs, detect anomalies, and optimize system performance. Voice control or facial recognition modules could also be incorporated to improve accessibility for differently abled users.

The hygiene-ensured smart door locking system represents an innovative step toward modernizing restroom management through automation and intelligent control. Its combination of IR sensors, servo motors, Arduino microcontroller, emergency alert

features, and real-time display ensures a touch-free, safe, and user-friendly experience. The system's simplicity, affordability, and scalability make it a practical choice for public and private infrastructures alike. By addressing hygiene, safety, and efficiency simultaneously, this design not only supports current public health needs but also aligns with the vision of sustainable and technologically advanced smart environments.

1.2 EMBEDDED SYSTEM

An embedded system is one kind of a computer system mainly designed to perform several tasks like to access, process, and store and also control the data in various electronics-based systems. Embedded systems are a combination of hardware and software where software is usually known as firmware that is embedded into the hardware. One of its most important characteristics of these systems is, it gives the o/p within the time limits. Embedded systems support to make the work more perfect and convenient. So, we frequently use embedded systems in simple and complex devices too. The applications of embedded systems mainly involve in our real life for several devices like microwave, calculators, TV remote control, home security and neighborhood traffic control systems, etc.

An embedded system is integration of hardware and software, the software used in the embedded system is set of instructions which are termed as a program. The microprocessors or microcontrollers used in the hardware circuits of embedded systems are programmed to perform specific tasks by following the set of instructions. These programs are primarily written using any programming software like Proteus or Lab-view using any programming languages such as C or C++ or embedded C. Then, the program is dumped into the microprocessors or microcontrollers that are used in the embedded system circuits.

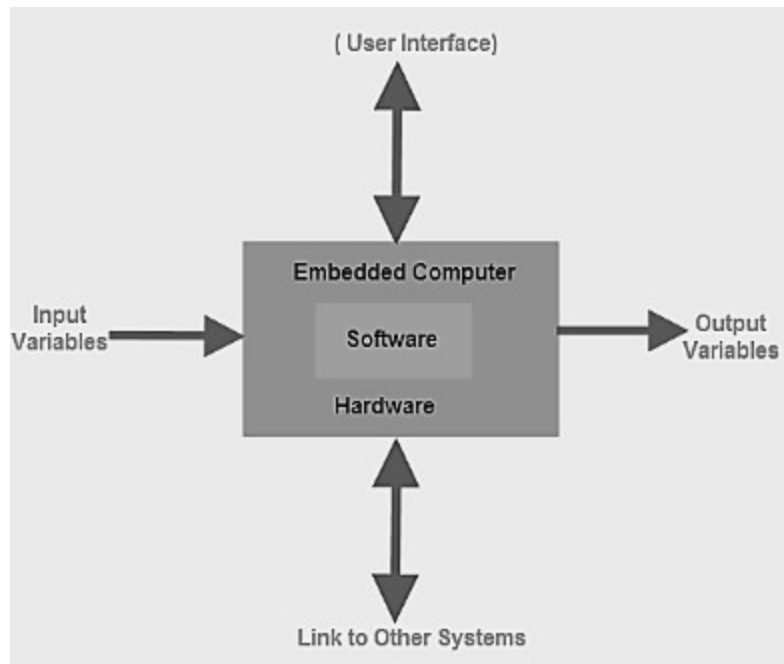


Fig 2: EMBEDDED SYSTEM

1.3 EMBEDDED SYSTEM CLASSIFICATION

Embedded systems are primarily classified into different types based on complexity of hardware & software and microcontroller (8 or 16 or 32-bit). Thus, based on the performance of the microcontroller, embedded systems are classified into three types such as:

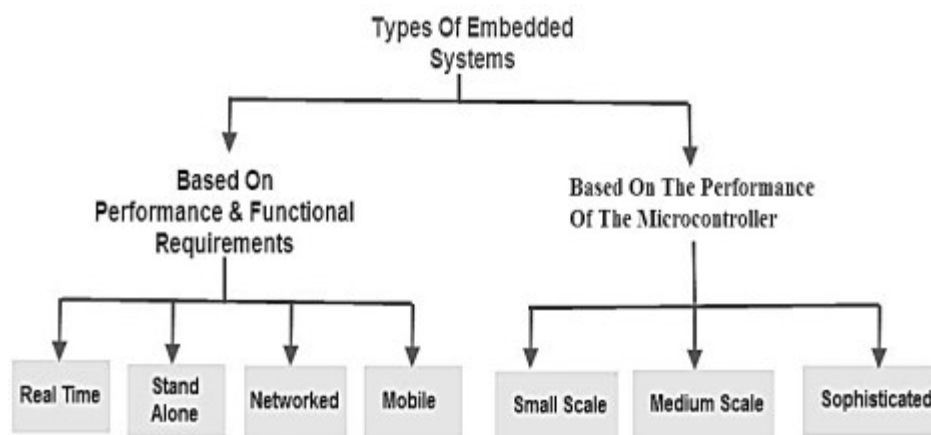


Fig 3: CLASSIFICATION

- Small scale embedded systems

- Medium scale embedded systems
- Sophisticated embedded systems

Further, based on performance and functional requirements of the system embedded system classified into four types such as:

- Standalone embedded systems
- Networked embedded systems
- Mobile embedded systems
- Real time embedded systems

Embedded System Hardware

An embedded system uses a hardware platform to perform the operation. Hardware of the embedded system is assembled with a microprocessor/microcontroller. It has the elements such as input/output interfaces, memory, user interface and the display unit. Generally, an embedded system comprises of the following

- Power Supply
- Memory
- Processor
- Timers
- Output/Output circuits
- Serial communication ports
- SASC (System application specific circuits)

Embedded System Software

The software of an embedded system is written to execute a particular function. It is normally written in a high-level setup and then compiled down to offer code that can be stuck within a non-volatile memory in the hardware. Embedded system software is intended to keep in view of the following three limits

- Convenience of system memory
- Convenience of processor's speed
- When the embedded system runs constantly, there is a necessity to limit power dissipation for actions like run, stop and wake up.

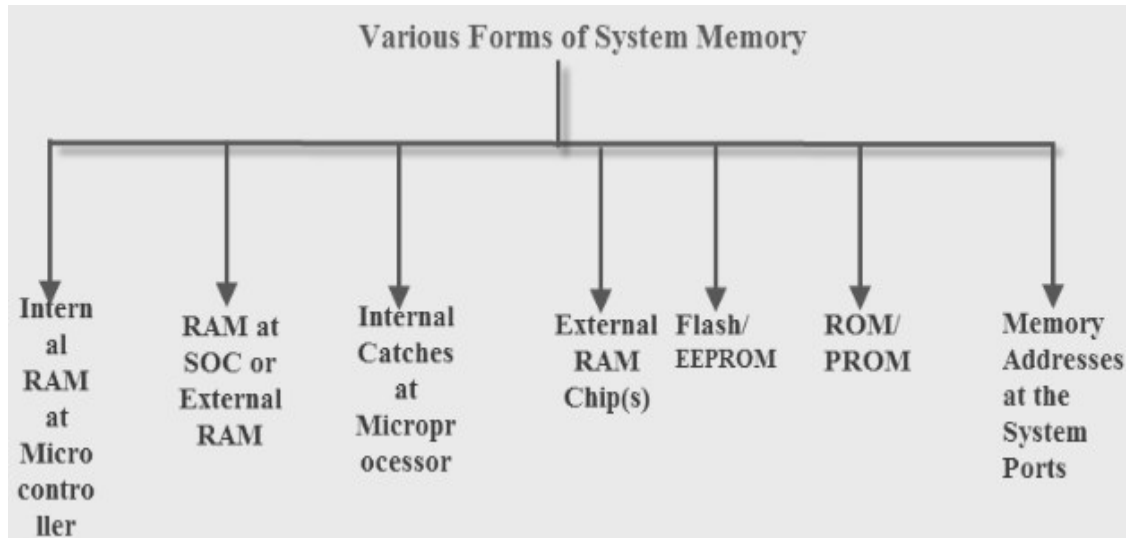


Fig 4: SYSTEM MEMORY

RTOS (Real Time Operating System)

A system which is essential to finish its task and send its service on time, then only it said to be a real time operating system. RTOS controls the application software and affords a device to allow the processor run. It is responsible for managing the different hardware resources of a personal computer and also host applications which run on the PC.

This operating system is specially designed to run various applications with an exact timing and a huge amount of consistency. Particularly, this can be significant in measurement & industrial automation systems where a delay of a program could cause a safety hazard.

Memory and Processors

The different kinds of processors used in an embedded system include Digital Signal Processor (DSP), microprocessor, RISC processor, microcontroller, ASSP processor, ASIP processor, and ARM processor. The different types of memories of an embedded system are given in the below chart.

Embedded System Characteristics

- Generally, an embedded system executes a particular operation and does the similar continually. For instance: A pager is constantly functioning as a pager.
- All the computing systems have limitations on design metrics, but those can be especially tight. Design metric is a measure of an execution features like size, power, cost and also performance.
- It must perform fast enough and consume less power to increase battery life.
- Several embedded systems should constantly react to changes in the system and also calculate particular results in real time without any delay. For instance, a car cruise controller; it continuously displays and responds to speed & brake sensors. It must calculate acceleration/de-accelerations frequently in a limited time; a delayed computation can consequence in let-down to control the car.
- It must be based on a microcontroller or microprocessor based.
- It must require a memory, as its software generally inserts in ROM. It does not require any secondary memories in the PC.
- It must need connected peripherals to attach input & output devices.
- An Embedded system is inbuilt with hardware and software where the hardware is used for security and performance and Software is used for more flexibility and features.

1.4 Embedded System Applications

The applications of an embedded system basics include smart cards, computer networking, satellites, telecommunications, digital consumer electronics, missiles, etc.



Fig 5: APPLICATIONS

- Embedded systems in automobiles include motor control, cruise control, body safety, engine safety, robotics in an assembly line, car multimedia, car entertainment, E-com access, mobiles etc.
- Embedded systems in telecommunications include networking, mobile computing, and wireless communications, etc.
- Embedded systems in smart cards include banking, telephone and security systems.

12 Embedded Systems in satellites and missiles include defense, communication, and aerospace.

CHAPTER 2

LITERATURE SURVEY

2.1 IoT-Enabled Smart Restroom and Septic Tank Hygiene Management System

Author: K. Sheikdavood et al.

Year: 2025

Description:

This paper presents an IoT-enabled smart restroom management system designed to enhance hygiene in both restrooms and associated septic tanks. The study emphasizes real-time monitoring of user presence, environmental parameters such as humidity and odor, and waste levels, with data transmitted to a central dashboard. The system integrates various sensors connected to an IoT framework, which triggers automated cleaning cycles, flush mechanisms, and maintenance alerts to ensure optimal hygiene standards. By leveraging cloud-based storage and data analytics, facility managers can proactively monitor restroom conditions and anticipate maintenance requirements, reducing service disruptions. The system also prioritizes water efficiency by controlling flush volumes based on usage patterns. The authors highlight that IoT integration provides a scalable and low-maintenance solution for high-traffic public restrooms. Security and data privacy concerns are addressed by implementing encrypted communication protocols between sensors and the cloud. The system's modular design allows retrofitting in existing facilities without significant structural modifications. Practical applications of this system include airports, hospitals, and large commercial complexes, where hygiene is critical for user safety. Experimental results demonstrate consistent detection accuracy and automated operation reliability, showing a significant reduction in water wastage and manual maintenance efforts. Overall, this study illustrates the feasibility and efficiency of integrating IoT with restroom management, setting a benchmark for hygiene optimization in modern smart infrastructure solutions.

2.2 Automated Toilet Cleaning System

Author: R. Vidyalakshmi et al.

Year: 2024

Description:

The paper focuses on the development of a fully automated toilet cleaning system, aimed at reducing human contact and improving hygiene. The system utilizes a combination of robotic actuators, sensors, and programmable control mechanisms to detect usage and initiate cleaning cycles. Sensors monitor occupancy, moisture levels, and residue presence to determine the timing and intensity of cleaning operations. The automated cleaning process involves coordinated movement of brushes, sprayers, and disinfectant dispensers to cover all internal surfaces effectively. The study emphasizes the importance of minimizing chemical usage while ensuring thorough disinfection, contributing to environmental sustainability. The system's control logic ensures that cleaning is performed only when the restroom is unoccupied, enhancing safety. The paper also highlights energy efficiency, as the robotic components are activated only when necessary. Simulation and experimental results indicate a substantial reduction in manual cleaning requirements and improved hygiene standards, making it suitable for hospitals, public facilities, and high-traffic areas. The system's adaptability is also discussed, with potential integration into IoT frameworks for remote monitoring and predictive maintenance. The study underscores the role of automation in reducing labor dependency while maintaining consistent sanitation standards. Challenges addressed include sensor calibration, obstacle detection, and coordination between mechanical actuators. Overall, this work demonstrates how robotics and automation can transform traditional restroom maintenance, providing a safer, cleaner, and more efficient hygiene management solution.

2.3 Optimizing Restroom Resources: A Smart Toilet Paper Dispensing System using Reinforcement Learning Algorithm

Author: D. Sasikala et al.

Year: 2024

Description:

This study introduces a smart toilet paper dispensing system that leverages reinforcement learning algorithms to optimize resource usage in restrooms. The system monitors user behavior patterns, toilet paper consumption rates, and usage frequency to predict and control dispensing amounts. Infrared and pressure sensors detect user interaction, allowing the algorithm to adjust the dispensing mechanism dynamically. By implementing a learning-based approach, the system continuously refines its decisions, reducing wastage and ensuring sufficient supply for users. The authors discuss the importance of balancing user convenience with sustainability objectives, emphasizing minimal environmental impact through reduced paper consumption. The smart system can be integrated into public and commercial restrooms to monitor inventory levels and trigger alerts when refills are required. The study also explores cost efficiency, noting potential savings from decreased paper usage and reduced maintenance frequency. Experimental simulations demonstrate that reinforcement learning significantly improves resource allocation compared to traditional time-based or fixed dispensing systems. The paper highlights the potential for integrating this system with IoT-enabled dashboards, allowing remote monitoring and predictive maintenance. Challenges include sensor sensitivity calibration, user variability, and ensuring robust algorithm performance under diverse conditions. Overall, this work provides a novel approach to resource optimization in restrooms, combining intelligent automation, sustainability, and user-focused design to create a smart and environmentally conscious facility management system.

2.4 Restroom: A Smart Restroom Cyberinfrastructure for Elderly People

Author: M. M. Rahman et al.

Year: 2022

Description:

The iRestroom project focuses on creating a smart restroom environment tailored specifically for elderly users, addressing accessibility, hygiene, and safety concerns. The system integrates a combination of motion sensors, automated door controls, and intelligent flush mechanisms to provide a user-friendly experience for seniors. The cyberinfrastructure monitors occupancy, environmental conditions, and usage patterns, while a central processing unit coordinates door operation, flushing, and emergency notifications. A major contribution of the study is the emphasis on fall prevention and assistance in critical situations through emergency alert systems connected to caregivers or facility managers. The system is designed with intuitive interfaces, ensuring that users with limited mobility can operate it independently. The study evaluates the system's effectiveness through trials involving elderly participants, demonstrating improved safety, accessibility, and hygiene. Data analytics is employed to monitor system performance, user behavior, and maintenance requirements. The integration of smart sensors and real-time monitoring provides a proactive approach to restroom management, ensuring cleanliness, user safety, and convenience. The paper also highlights the importance of designing adaptive systems that cater to the specific needs of vulnerable populations, combining automation with accessibility features. Overall, iRestroom exemplifies the integration of cyberinfrastructure, IoT, and user-centric design to create intelligent and safe restroom facilities for the elderly.

2.5 Robotic Restroom Hygiene Solutions with IoT and Recurrent Neural Networks (RNNs)

Author: K. Nasreen et al.

Year: 2024

Description:

This research explores the integration of IoT-enabled robotic systems with artificial intelligence, specifically Recurrent Neural Networks (RNNs), for restroom hygiene management. The system employs IoT sensors to detect usage patterns, environmental conditions, and cleaning requirements, while RNNs predict optimal cleaning schedules and resource allocation. By analyzing historical data, the neural network can anticipate peak usage periods, allowing the robotic cleaning modules to operate efficiently without interfering with users. The robotic components include automated scrubbers, disinfectant sprayers, and intelligent flush controls. A key feature is the system's predictive maintenance capability, which reduces downtime and ensures consistent hygiene. The study demonstrates that combining IoT data collection with RNN-based predictive algorithms improves efficiency and reduces operational costs compared to reactive cleaning approaches. Challenges addressed include sensor calibration, robot navigation in confined spaces, and ensuring real-time responsiveness to user interactions. Results indicate improved hygiene standards, minimized water and chemical usage, and enhanced overall restroom cleanliness. The paper highlights the potential for deploying intelligent robotic systems in hospitals, high-traffic public spaces, and commercial facilities, illustrating how AI-driven automation can transform traditional maintenance practices into proactive, data-driven hygiene solutions.

2.6 An IoT-Based Hygiene Monitoring System in the Restroom

Author: F. I. Azman et al.

Year: 2025

Description:

This paper presents an IoT-based restroom hygiene monitoring system designed to automate and enhance cleanliness while providing real-time data for facility management. The system integrates a network of sensors that detect user presence, restroom occupancy, environmental conditions (humidity, odor, and moisture), and flushing events. All data is transmitted to a cloud-based platform for centralized monitoring, enabling facility managers to analyze usage patterns, identify maintenance needs, and optimize cleaning schedules. Automated alerts are generated for conditions requiring immediate attention, such as overflowing waste bins or high humidity levels that may encourage microbial growth. The study highlights energy efficiency by activating sensors and devices only when required, reducing operational costs. The system also improves user safety by incorporating emergency buttons linked to notifications for maintenance staff or security personnel. Detailed experimental evaluations demonstrate high accuracy in occupancy detection and sensor responsiveness, even under high-traffic conditions. The modular design allows easy integration into existing restroom infrastructures without major structural modifications. Furthermore, the paper emphasizes scalability and adaptability, allowing expansion to multiple restroom units or integration with other smart building systems. By combining IoT sensors, cloud computing, and automated alert mechanisms, this system significantly reduces manual monitoring requirements while ensuring a hygienic, safe, and user-friendly restroom environment. Overall, the work illustrates the practical implementation of smart monitoring systems in maintaining hygiene standards in public and healthcare facilities.

2.7 Smart Toilet Monitoring System Using IoT For Hygiene And Health Diagnostics

Author: N.B. Mahesh Kumar

Year: 2025

Description:

This study introduces a smart toilet system leveraging IoT technology for both hygiene management and health diagnostics. The system uses multiple sensors to detect user presence, measure toilet usage, and monitor environmental conditions such as moisture and odor. The data collected is analyzed to ensure timely cleaning, prevent contamination, and maintain optimal hygiene levels. A significant contribution of this work is the health diagnostic capability; the system can monitor certain human biometrics and provide alerts if abnormal conditions are detected, such as signs of dehydration or irregular urination frequency. These insights are transmitted securely to cloud servers, allowing health professionals or facility managers to access anonymized health trends and maintenance data. Automated flushing mechanisms and touchless operation reduce human contact, enhancing hygiene standards. The system also features an LCD display to show real-time operational status, including flush completion, occupancy, and maintenance alerts. Experimental evaluations indicate the system reliably detects occupancy, triggers flushing correctly, and reduces manual cleaning requirements, while also offering preliminary health insights for users. The modular and scalable design ensures adaptability across different restroom types, from public spaces to healthcare centers. By integrating IoT-based monitoring, automated hygiene management, and health diagnostic capabilities, this system offers a comprehensive approach to restroom hygiene and proactive health management.

2.8 An Arduino-Based Smart Digital Flushable Toilet for Low-Income Communities

Author: W. Kabaso et al.

Year: 2023

Description:

This paper presents a cost-effective smart digital flushable toilet designed specifically for low-income communities, emphasizing hygiene, affordability, and ease of maintenance. The system uses an Arduino microcontroller to coordinate sensor-based detection, automatic flushing, and real-time operational feedback. Infrared sensors detect user presence, triggering the flush system only after usage, which ensures water efficiency and reduces contamination risks. The design prioritizes simplicity and durability, allowing deployment in areas with limited technical resources. The flush system operates automatically with calibrated water volumes to prevent wastage while maintaining hygiene standards. The paper highlights the socio-economic impact, demonstrating that automated sanitation can be made accessible to communities with limited infrastructure, enhancing public health outcomes. Implementation studies indicate reliable sensor performance, responsive flush mechanisms, and minimal maintenance requirements. Additionally, the modular nature of the system allows easy replacement of components, further reducing operational costs. Environmental benefits include reduced water consumption and improved sanitation hygiene, which is critical in regions prone to waterborne diseases. Overall, this study provides a practical model for delivering automated sanitation solutions to underserved populations, combining technology, cost efficiency, and public health awareness.

2.9 Arduino-based Automated Washroom Sanitizing System

Author: M. Anto et al.

Year: 2020

Description:

This study focuses on the design and implementation of an Arduino-based automated washroom sanitizing system aimed at minimizing manual cleaning efforts while maintaining hygiene. The system utilizes infrared sensors to detect user presence, initiating automated sanitization processes, including spray-based disinfectant application, surface cleaning, and automated flush control. A central Arduino microcontroller coordinates sensor inputs and actuator outputs, ensuring precise timing and sequencing of cleaning operations. Safety mechanisms are implemented to prevent system activation while the restroom is occupied, protecting users from exposure to cleaning chemicals. The study also addresses water and chemical efficiency, optimizing the amount used per cleaning cycle to reduce wastage and operational costs. Experimental testing shows the system reliably detects occupancy, triggers sanitization, and maintains consistent cleaning coverage across high-traffic areas. Integration with an LCD display provides real-time status updates on sanitization cycles, flush completion, and occupancy detection. The research highlights the adaptability of Arduino-based platforms for developing affordable and efficient hygiene solutions in public restrooms. Overall, this work demonstrates that low-cost microcontroller-based systems can automate restroom maintenance effectively, improving hygiene, reducing labor dependency, and ensuring a safer environment for users.

2.10 Scheduling and Predictive Maintenance for Smart Toilet

Author: Lokman et al.

Year: 2023

Description:

This paper proposes a scheduling and predictive maintenance framework for smart toilets to enhance operational efficiency and reduce downtime. The system employs IoT sensors to monitor usage frequency, flush performance, water levels, and environmental conditions such as humidity and cleanliness. Data collected is processed to predict maintenance requirements, allowing proactive intervention before system failures occur. The study highlights the integration of scheduling algorithms to optimize cleaning cycles based on restroom occupancy patterns and peak usage hours. Predictive maintenance reduces water wastage, ensures uninterrupted service, and maintains hygiene standards. The system also incorporates automated alerts and notifications for maintenance personnel, improving responsiveness to system anomalies or failures. Experimental validation demonstrates accurate detection of usage patterns, timely initiation of cleaning cycles, and reduced manual intervention. The study emphasizes scalability and adaptability, allowing deployment in multi-unit public restroom facilities or hospitals. By combining IoT-based monitoring, scheduling algorithms, and predictive analytics, the system provides a data-driven approach to restroom hygiene management. Overall, this research illustrates how smart systems can efficiently maintain cleanliness, enhance user satisfaction, and optimize resource utilization while reducing operational costs and maintenance complexity.

CHAPTER 3

EXISTING SYSTEM

In traditional restroom management systems, the process of maintaining hygiene, safety, and privacy largely depends on manual operations. The existing restroom systems typically involve mechanical door locks, manual flushing mechanisms, and user-operated switches for lighting or ventilation. Although such systems are functional and widely used, they suffer from several drawbacks related to hygiene, efficiency, and user safety. Manual contact with restroom components, especially in high-traffic public areas, increases the chances of spreading infectious diseases. As human touch remains the primary mode of contamination transfer, the need for automation in such environments has become increasingly significant. The existing systems lack the technological integration necessary to provide contactless operation, real-time monitoring, or automated hygiene enforcement, which are essential for modern public facilities.

Most conventional restrooms rely on mechanical door locking systems, where users must physically push, pull, or turn a handle to open and close the door. These mechanical systems are simple, low-cost, and easy to install but require continuous physical contact. Over time, door handles become high-touch surfaces, harboring bacteria and viruses from multiple users. This problem is further compounded in hospitals, airports, schools, and shopping malls, where large crowds use restrooms daily. Although some modern facilities have introduced foot-operated or sensor-based door systems, these are often limited to high-end installations and lack full integration with other restroom functions such as flushing or emergency alerts. Therefore, existing systems fail to provide a fully automated and hygienic environment suitable for large-scale deployment.

Another critical component of traditional restroom infrastructure is the manual flushing mechanism. Most restrooms utilize either a push-button or lever-based flush system, which again requires physical contact. Some facilities have implemented sensor-based automatic flush systems, but these are usually standalone units that function independently without coordination with other systems like door locking or occupancy detection. This lack of integration leads to inefficiencies in water usage and does not guarantee proper hygiene control between consecutive users. Moreover, in many public

restrooms, malfunctioning flush systems or improper usage lead to unhygienic conditions, causing odor and discomfort for the next user. Manual flushing systems are also prone to human negligence—users may forget or deliberately avoid flushing after use, further compromising sanitation standards.

The safety aspect of existing restroom systems is also minimal. In traditional setups, there is rarely any mechanism for emergency assistance in case of accidents, fainting, or other health-related incidents inside the restroom. Users may be trapped inside due to mechanical failure or health distress without a way to signal for help. This issue becomes especially concerning in hospitals, elderly care centers, or disabled-friendly restrooms, where users may require quick assistance. The absence of an emergency alert feature in most conventional designs poses a serious risk to user safety and limits the usability of these facilities for vulnerable individuals.

Additionally, the monitoring and feedback capabilities in existing systems are highly limited. Traditional restrooms do not provide real-time information about occupancy, cleanliness, or maintenance status. Facility managers often rely on manual inspection to determine whether restrooms are functioning properly or need cleaning. This manual monitoring approach is inefficient, labor-intensive, and prone to delays, especially in large public infrastructures. Without automation, it becomes challenging to maintain consistent hygiene levels or track resource usage efficiently. As a result, even when automated flushing or lighting systems are implemented, they often operate in isolation without centralized control or synchronization between components.

From a technological perspective, the absence of microcontroller-based integration is one of the most significant limitations of existing systems. The lack of a central control unit means that actions such as door operation, flushing, and safety alerts cannot be coordinated intelligently. For instance, if a flush sensor is triggered before the user exits, the door may open prematurely, compromising hygiene and privacy. Similarly, without synchronized control, a malfunction in one component can disrupt the overall functionality. This lack of coordination reduces the overall efficiency and reliability of restroom automation, highlighting the need for an integrated approach that ensures sequential operation of all components.

Energy and resource management also present challenges in the existing system. Manual lighting and flushing systems often lead to unnecessary power and water wastage. Lights are frequently left on even when the restroom is unoccupied, and excessive flushing wastes significant amounts of water daily. While motion-sensor lighting solutions have partially addressed this issue, they do not integrate with occupancy detection mechanisms that govern other restroom functions. Consequently, energy optimization remains limited in traditional setups. In addition, maintenance costs increase due to the frequent need for manual inspections, repairs, and cleaning interventions that could otherwise be automated through real-time monitoring.

Hygiene maintenance in existing systems is further constrained by human behavior and inconsistency. Public users often fail to follow hygiene protocols, such as washing hands or ensuring the cleanliness of the facility after use. This behavior undermines manual cleaning efforts and contributes to an unhygienic environment. Moreover, during epidemic or pandemic situations, such as the COVID-19 outbreak, the limitations of manual systems became even more evident. The need for frequent sanitization, minimal touchpoints, and social distancing revealed that traditional restrooms are not designed to handle public health challenges efficiently. Contactless systems were no longer considered a luxury but a necessity for preventing the spread of infection in shared spaces.

In terms of cost and scalability, while traditional systems are inexpensive to install, they become costly to maintain in the long term due to frequent cleaning requirements, mechanical wear, and user negligence. Additionally, upgrading such systems to include automation features is difficult without redesigning the existing infrastructure. The lack of modularity makes integration with modern technologies—such as IoT-based monitoring, automated cleaning, or smart access control—highly challenging. This results in outdated facilities that cannot adapt to evolving hygiene and safety standards.

From the user experience perspective, traditional restrooms often fail to provide convenience, accessibility, and assurance of cleanliness. Manual locks can be difficult to operate for children, elderly individuals, or people with disabilities. Similarly, the absence of automated feedback systems means users have no way of knowing whether the restroom

is clean, occupied, or available for use. The lack of intelligent indicators leads to confusion and inefficiency in managing restroom queues in high-traffic environments.

The existing restroom systems, while functional, are outdated in terms of hygiene, safety, and automation. They depend heavily on manual interaction, lack coordination between operations, and provide no real-time monitoring or emergency response features. The limitations of these systems emphasize the urgent need for a comprehensive solution that integrates sensing, control, and automation into a single framework. The hygiene-ensured smart door locking system is proposed to overcome these shortcomings by introducing an intelligent, touchless mechanism that ensures hygiene, privacy, safety, and operational efficiency. Through microcontroller-based control, synchronized operation, and sensor integration, the new system aims to transform traditional restrooms into smart, hygienic environments suitable for modern public and private infrastructures.

3.1 DISADVANTAGES

- **Lack of Hygiene Control:** Manual operation of doors, flush handles, and taps leads to frequent human contact, increasing the risk of germ transmission and contamination.
- **Absence of Automation:** Existing systems rely on full manual operation without intelligent control or automatic sequencing of restroom functions.
- **No Integration Between Components:** Independent operation of flushing, lighting, and door systems causes inefficiency and lack of coordination.
- **Manual Flushing Mechanism:** Users must operate flush handles or buttons manually, leading to hygiene issues and inconsistent cleaning.
- **Negligence in User Behavior:** Users often forget or avoid flushing, creating unhygienic and unpleasant conditions for subsequent users.
- **No Emergency Assistance Feature:** Traditional systems lack emergency alert mechanisms to help users in distress or during mechanical failures.
- **Inadequate Privacy Management:** Manual locks and uncoordinated mechanisms may compromise privacy during restroom usage.
- **High Risk of Cross-Contamination:** Door handles and flush buttons become high-touch infection points, particularly in crowded public facilities.

CHAPTER 4

PROPOSED SYSTEM

The proposed Hygiene Ensured Smart Door Locking System is an intelligent automation solution designed to enhance hygiene, safety, and convenience within restroom environments. It addresses the shortcomings of traditional restroom systems by integrating sensor technology, microcontroller-based control, and automated actuation mechanisms. The system minimizes physical contact, maintains sequential control of restroom operations, and ensures a clean, private, and safe user experience. The design is centered around the Arduino microcontroller, which acts as the central processing unit responsible for coordinating sensor inputs, processing logical decisions, and controlling output devices such as the servo motor, buzzer, and LCD display. The proposed system not only promotes hygiene but also enhances user safety and operational efficiency, making it ideal for deployment in public restrooms, hospitals, schools, offices, and smart infrastructure environments.

The system's architecture is built on the concept of contactless operation, where each action—entry, usage, flushing, and exit—is automated through sensor detection and programmed logic. Two Infrared (IR) sensors form the foundation of the sensing mechanism. The first IR sensor is installed near the entrance to detect the presence of an approaching person. When this sensor is triggered, the Arduino sends a command to the servo motor to rotate to the open position (typically 90°), allowing entry. After a short delay, the door automatically closes to ensure privacy. This eliminates the need for manual door handling, significantly reducing the risk of contamination. Once the user enters, the door remains locked, ensuring that no one else can access the restroom until the sequence of operations is completed.

The second IR sensor, positioned near the flushing unit, is responsible for detecting user activity within the restroom. When it senses that the toilet has been used, the Arduino activates the flushing mechanism. This automatic process ensures that hygiene is maintained after every use, without requiring manual intervention. The system also maintains a logical sequence—ensuring that the door remains locked during the flushing

process and only reopens after it is complete. This sequencing prevents premature access and ensures proper sanitation before the next user enters. The synchronization between the sensors, servo motor, and flushing process represents a key improvement over existing systems, enabling a fully coordinated automation cycle.

An additional feature of the proposed design is the inclusion of an emergency alert mechanism. Public restrooms often cater to a diverse range of users, including the elderly, disabled, or individuals with medical conditions who might face unexpected distress. To ensure user safety, an emergency switch is installed inside the restroom. When pressed, it activates a buzzer, which generates a loud alert to notify nearby staff or maintenance personnel. This immediate notification system provides users with a sense of security and ensures rapid response during emergencies. Such safety integration transforms the restroom from a passive facility into a responsive and intelligent environment capable of handling critical situations autonomously.

To provide transparency and real-time monitoring, a 16×2 Liquid Crystal Display (LCD) is incorporated into the system. This display serves as a user interface, showing operational information such as door status (open or closed), flush activity, person detection, and emergency alerts. For example, the screen might display “Door Locked – In Use” or “Flushing in Progress,” giving users and maintenance personnel clear status updates. This feature not only improves usability but also facilitates maintenance by allowing technicians to identify system faults or operational stages easily. The LCD can also be adapted in future iterations to support multilingual messages or graphical icons for enhanced user comprehension.

At the heart of the proposed system, the Arduino microcontroller performs real-time decision-making based on input signals from the sensors and switch. The Arduino’s program is written in C/C++ using a defined logic structure that controls the timing and interaction between devices. The control logic ensures smooth transitions between each operational phase—detecting entry, opening and closing the door, triggering the flush, and reopening after completion. For example, if the entrance sensor detects a person, the system checks whether the restroom is occupied or in a flushing cycle before unlocking the door.

This intelligent control flow prevents operational conflicts and ensures consistent performance under different conditions.

The hardware design is compact, cost-effective, and modular. The use of low-power components ensures that the system can operate efficiently even in continuous-use environments. The servo motor provides precise angular control for the door, while the IR sensors offer reliable motion detection unaffected by ambient light variations. All components are powered through the Arduino's regulated supply, simplifying installation and maintenance. Additionally, the wiring layout is designed to reduce complexity and improve durability, making the system suitable for both new installations and retrofitting in existing restroom structures.

From an operational standpoint, the proposed system offers a seamless and touch-free experience. The entire process—from entry to exit—is automated without requiring the user to touch any surface. The stepwise operation is as follows: The entrance IR sensor detects a person approaching. The Arduino signals the servo motor to open the door. After the person enters, the door closes automatically, ensuring privacy. The second IR sensor detects restroom usage and triggers the automatic flush. The door remains closed during the flushing process to maintain hygiene. After flushing, the door reopens, allowing the user to exit. In case of an emergency, pressing the switch activates the buzzer and alert system.

This structured and timed sequence minimizes errors, ensures proper hygiene maintenance, and optimizes the overall user experience. Unlike traditional systems, it guarantees that flushing always occurs after every use, eliminating the issue of unflushed toilets.

The advantages of the proposed system extend beyond hygiene automation. It significantly reduces human involvement, thereby lowering the chances of cross-contamination. The use of automation also improves the consistency of restroom maintenance by enforcing fixed sequences for each cycle. Additionally, the system promotes water conservation by automating flush duration and ensuring timely operation. Safety is enhanced through the emergency alert mechanism, and privacy is maintained through controlled door actuation. Furthermore, the LCD display ensures clarity and

usability for both users and maintenance personnel, making it a comprehensive and user-centric solution.

The proposed system can also serve as a foundation for future enhancements. For instance, Internet of Things (IoT) integration could allow remote monitoring of restroom usage, water consumption, and maintenance alerts. Facility managers could receive notifications about system performance or emergency activations in real time. The design can also be expanded to include features like automatic air fresheners, temperature or humidity sensors, and voice-controlled accessibility for differently abled users. Additionally, solar-powered operation could make the system more sustainable for outdoor or off-grid applications.

The proposed hygiene-ensured smart door locking system provides a modern, efficient, and user-friendly solution to the challenges faced by traditional restroom systems. By combining sensor technology, Arduino-based control, and intelligent automation, it ensures hygiene, safety, and comfort without compromising on affordability or simplicity. The system's modular design and real-time operational feedback make it adaptable for various environments, paving the way for future innovations in smart sanitation. Ultimately, this proposed solution exemplifies how automation can be leveraged to create cleaner, safer, and more sustainable public facilities.

4.1 PROPOSED SYSTEM ADVANTAGES

- **Touch-Free Operation:** Completely eliminates physical contact with doors and flush mechanisms, thereby reducing the spread of germs and maintaining high hygiene standards.
- **Automated Door Control:** Servo motor-based door automation provides smooth opening and closing operations, ensuring convenience and privacy without manual effort.
- **Sequential System Logic:** Ensures that entry, usage, flushing, and exit occur in a logical, time-coordinated sequence, minimizing user confusion and system errors.

- **Enhanced Hygiene Maintenance:** Automatic flushing after every use guarantees that the toilet remains clean for the next user, preventing odor and contamination.
- **Emergency Safety Mechanism:** Equipped with an emergency switch that triggers a buzzer alarm, allowing immediate response in case of accidents or distress.
- **Real-Time Status Display:** LCD display provides instant updates such as “Door Locked,” “Flushing in Progress,” or “Available,” improving user experience and operational transparency.
- **Energy-Efficient Design:** Uses low-power components like IR sensors and Arduino, ensuring minimal electricity consumption and sustainable operation.
- **Improved User Privacy:** Automatic locking and unlocking maintain user privacy and prevent intrusions during restroom use.
- **Reduced Human Intervention:** Minimizes the need for cleaning staff to operate or monitor the restroom manually, enhancing efficiency and reducing labor costs.
- **Water Conservation:** The automated flush ensures optimized water usage, preventing wastage due to user negligence or repeated manual flushing.
- **Cost-Effective Implementation:** Built using affordable, easily available components (Arduino, IR sensors, servo motor), making it suitable for large-scale deployment.
- **Easy Maintenance and Upgradation:** Modular design allows for quick repair or replacement of individual parts without complex rewiring or system overhaul.
- **High Reliability and Accuracy:** IR sensors ensure precise detection of user presence and movement, leading to consistent and reliable system performance.
- **Hygienic Environment Assurance:** By eliminating manual contact and ensuring automatic cleaning, it maintains a consistently hygienic restroom environment.
- **User-Friendly Design:** The stepwise automation process is intuitive and requires no user learning curve, making it accessible for people of all ages.
- **Enhanced Safety for Vulnerable Users:** Emergency alert ensures protection for children, elderly individuals, and differently abled users who may face sudden issues.

- **Adaptable for Smart Infrastructure:** Can be easily integrated into modern smart buildings, hospitals, airports, and corporate offices as part of IoT-based hygiene management systems.
- **Reduced Cross-Contamination:** Since no handles, switches, or buttons are touched, the risk of spreading infectious diseases through surfaces is drastically minimized.
- **Low Installation Complexity:** Compact circuit and simple wiring make installation straightforward in both new and existing restrooms.
- **Scalability and Future Expansion:** The design can be extended to include IoT monitoring, voice commands, or smart energy systems for advanced automation.

CHAPTER 5

SYSTEM SPECIFICATION

5.1 HARDWARE REQUIREMENTS

5.1.1. ARDUINO UNO

The Arduino UNO is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 Digital pins, 6 Analog pins, and programmable with the Arduino IDE (Integrated Development Environment) via a type B USB cable. It can be powered by a USB cable or by an external 9 volt battery, though it accepts voltages between 7 and 20 volts. It is also similar to the Arduino Nano and Leonardo. The hardware reference design is distributed under a Creative Commons Attribution Share-Alike 2.5 license and is available on the Arduino website. Layout and production files for some versions of the hardware are also available. "Uno" means one in Italian and was chosen to mark the release of Arduino Software (IDE) 1.0. The Uno board and version 1.0 of Arduino Software (IDE) were the reference versions of Arduino, now evolved to newer releases. The Uno board is the first in a series of USB Arduino boards, and the reference model for the Arduino platform. The ATmega328 on the Arduino Uno comes pre-programmed with a boot loader that allows uploading new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol. The Uno also differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it uses the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter



Fig 6: ARDUINO UNO

The Arduino project started at the Interaction Design Institute Ivrea (IDII) in Ivrea, Italy. At that time, the students used a BASIC Stamp microcontroller at a cost of \$100, a considerable expense for many students. In 2003 Hernando Barragán created the development platform Wiring as a Master's thesis project at IDII, under the supervision of Massimo Banzi and Casey Reas, who are known for work on the Processing language. The project goal was to create simple, low-cost tools for creating digital projects by non-engineers. The Wiring platform consisted of a printed circuit board (PCB) with an ATmega168 microcontroller, an IDE based on Processing and library functions to easily program the microcontroller. In 2003, Massimo Banzi, with David Mellis, another IDII student, and David Cuartielles, added support for the cheaper ATmega8 microcontroller to Wiring. But instead of continuing the work on Wiring, they forked the project and renamed it Arduino. Early arduino boards used the FTDI USB-to-serial driver chip and an ATmega168. The Uno differed from all preceding boards by featuring the ATmega328P microcontroller and an ATmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter.

SPECIFICATION:

- ✓ Microcontroller: Microchip ATmega328P
- ✓ Operating Voltage: 5 Volt
- ✓ Input Voltage: 7 to 20 Volts
- ✓ Digital I/O Pins: 14 (of which 6 provide PWM output)
- ✓ Analog Input Pins: 6

- ✓ DC Current per I/O Pin: 20 mA
- ✓ DC Current for 3.3V Pin: 50 mA
- ✓ Flash Memory: 32 KB of which 0.5 KB used by boot loader
- ✓ SRAM: 2 KB
- ✓ EEPROM: 1 KB
- ✓ Clock Speed: 16 MHz
- ✓ Length: 68.6 mm
- ✓ Width: 53.4 mm
- ✓ Weight: 25 g

COMMUNICATION:

The Arduino/Genuino Uno has a number of facilities for communicating with a computer, another Arduino/Genuino board, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The 16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, an .inf file is required. The Arduino Software (IDE) includes a serial monitor which allows simple textual data to be sent to and from the board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1). A Software Serial library allows serial communication on any of the Uno's digital pins

PINS General Pin functions

- LED: There is a built-in LED driven by digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.
- VIN: The input voltage to the Arduino/Genuino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated

power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.

- **5V:** This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 20V), the USB connector (5V), or the VIN pin of the board (7-20V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage the board.
- **3V3:** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND:** Ground pins.
- **IOREF:** This pin on the Arduino/Genuino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin voltage and select the appropriate power source or enable voltage translators on the outputs to work with the 5V or 3.3V.
- **Reset:** Typically used to add a reset button to shields which block the one on the board.

Special Pin Functions:

Each of the 14 digital pins and 6 Analog pins on the Uno can be used as an input or output, using `pinMode()`, `digitalWrite()`, and `digitalRead()` functions. They operate at 5 volts. Each pin can provide or receive 20 mA as recommended operating condition and has an internal pull-up resistor (disconnected by default) of 20-50k ohm. A maximum of 40mA is the value that must not be exceeded on any I/O pin to avoid permanent damage to the microcontroller. The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though is it possible to change the upper end of their range using the AREF pin and the `analogReference()` function.

In addition, some pins have specialized functions:

- Serial / UART: pins 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- External Interrupts: pins 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- PWM (Pulse Width Modulation): 3, 5, 6, 9, 10, and 11 Can provide 8-bit PWM output with the `analogWrite()` function.
- SPI (Serial Peripheral Interface): 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication using the SPI library.
- TWI (Two Wire Interface) / I²C: A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library.
- AREF (Analog REference): Reference voltage for the analog inputs

5.1.2 LIQUID CRYSTAL DISPLAY (LCD):

A liquid crystal display (LCD) is a flat panel display, electronic visual display, or video display that uses the light modulating properties of liquid crystals. Liquid crystals do not emit light directly. LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images which can be displayed or hidden, such as preset words, digits, and 7-segment displays as in a digital clock. They use the same basic technology, except that arbitrary images are made up of a large number of small pixels, while other displays have larger elements. An LCD is a small low cost display. It is easy to interface with a micro-controller because of an embedded controller (the black blob on the back of the board). This controller is standard across many displays (HD 44780) which means many micro-controllers (including the Arduino) have libraries that make displaying messages as easy as a single line of code.

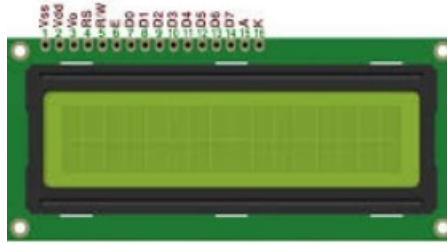


Fig 7: LCD display unit

LCDs are used in a wide range of applications including computer monitors, televisions, instrument panels, aircraft cockpit displays, and signage. They are common in consumer devices such as video players, gaming devices, clocks, watches, calculators, and telephones, and have replaced cathode ray tube (CRT) displays in most applications. They are available in a wider range of screen sizes than CRT and plasma displays, and since they do not use phosphors, they do not suffer image burn-in. LCDs are, however, susceptible to image persistence.

16X2 LCD SPECIFICATIONS

- ✓ Display Format: 16 characters per line, 2 lines total.
- ✓ Character Size: 5x8 pixels for standard characters.
- ✓ Dimensions: Approximately 80mm x 36mm x 13mm.
- ✓ Interface: Parallel (4-bit or 8-bit mode).
- ✓ Supply Voltage: Typically 5V DC.
- ✓ Current Consumption: Around 1.5 mA at 5V.
- ✓ Backlight: LED backlight (3.3V to 5V).
- ✓ Temperature Range: 0°C to 70°C operating, -20°C to 80°C storage.
- ✓ Response Time: Under 10 ms.
- ✓ Mounting: PCB or breadboard compatible.
- ✓ Character Set: Standard ASCII with custom character support.

5.1.3 POWER SUPPLY

The 7812 and 7805 voltage regulators are commonly used components to provide stable DC voltage outputs of +12V and +5V, respectively, from a higher input voltage source.



Fig 8: VOLTAGE REGULATOR MODULE

7812 Voltage Regulator:

Input Voltage: Typically requires an input voltage slightly higher than 12V (usually around 14-16V) to regulate effectively.

Output Voltage: Provides a stable +12V DC output.

Capacitors:

C1000/25: This likely refers to a capacitor with a capacitance of 1000 μ F and a voltage rating of 25V. This capacitor is typically placed on the input side (between input and ground) to stabilize the input voltage, reducing noise and providing a reservoir of charge to handle transient spikes.

C10/63: This could refer to a capacitor with a capacitance of 10 μ F and a voltage rating of 63V. This capacitor is usually placed on the output side (between output and ground) to stabilize the output voltage, filtering out any remaining noise and improving regulation.

Resistor: A resistor isn't typically used directly with the 7812 regulator in the same way as capacitors are, but it can be part of the circuit design for specific applications, such as in voltage dividers or as part of a feedback loop for stability.

7805 Voltage Regulator:

Input Voltage: Requires an input voltage typically around 7-25V (ideal 7-20V) to regulate effectively.

Output Voltage: Provides a stable +5V DC output.

Capacitors:

C1000/25: As with the 7812, this capacitor stabilizes the input voltage to the regulator.

C10/63: This capacitor stabilizes the output voltage of the regulator.

Resistor: Similar to the 7812, resistors are not directly part of the typical configuration but can be used in specific applications.

Circuit Considerations:

Decoupling Capacitors: These capacitors (like C1000/25 and C10/63) are crucial for filtering out noise and stabilizing the voltage levels, ensuring reliable operation of your circuit.

Heat Dissipation: Both regulators can generate heat, especially when dropping significant voltage. Adequate heat sinking may be necessary depending on the current drawn and the input-output voltage differential.

Current Requirements: Ensure that the regulators can supply enough current for your application. If higher currents are required, additional heat sinking and possibly parallel regulators may be needed.

In summary, the 7812 and 7805 voltage regulators, along with capacitors like C1000/25 and C10/63, form a basic yet effective setup for providing stable +12V and +5V outputs in electronic circuits, suitable for a wide range of applications from powering microcontrollers to analog circuits.

5.1.4 BUZZER



Fig 9: BUZZER

A buzzer is an electrical or mechanical device designed to produce a loud, sharp, and often continuous sound, typically used as an alert or signalling tool. Buzzers are commonly used in various applications where attention is needed quickly, such as in alarm systems, games, and industrial environments. In most cases, buzzers function through an oscillating or vibrating mechanism that generates sound waves, usually driven by electric currents. These devices come in different forms, including piezoelectric buzzers, which produce sound through the mechanical vibrations of a crystal, and electromagnetic buzzers, where a magnetic field induces vibrations to generate sound.

Buzzers are integral components in systems requiring real-time notifications or alerts. In educational or competitive settings, such as quiz shows or games, buzzers are used to allow participants to signal their intent to answer a question. In industrial settings, buzzers might warn workers of hazards, equipment malfunctions, or the need for immediate attention. With advancements in technology, buzzers can be integrated into IoT systems, connected devices, and automated control systems, providing both auditory and visual signals to improve user interaction and safety.

5.1.5 SERVO MOTOR:

A servo motor is a rotary actuator or linear actuator that allows for precise control of angular or linear position, velocity, and acceleration. It consists of a motor coupled to a

sensor for position feedback. A servo motor is commonly used in robotics, automation, and control systems due to its precision and reliability.

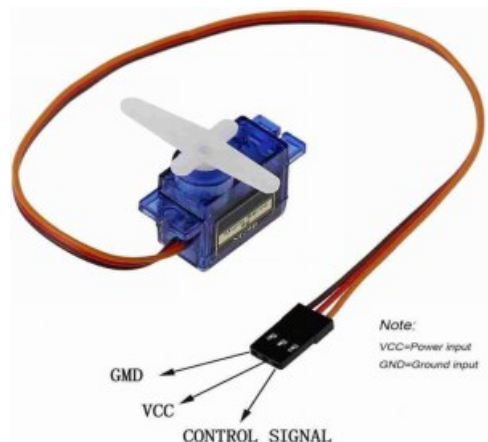


Fig 10: SERVO MOTOR

Principle of Operation:

The servo motor operates on the principle of closed-loop feedback. It receives a control signal that represents the desired output position and uses a feedback mechanism to compare this signal with the actual position. The motor adjusts until the actual position matches the desired position. This feedback loop allows for precise control of the motor's position.

Pin Configuration

Power (VCC):	5V
Ground (GND):	Ground of the power supply (0V)
Control (Signal):	Control the position of the servo motor.

Interfacing with Arduino Uno:

To interface a servo motor with an Arduino Uno, begin by connecting the servo motor's VCC pin to the Arduino's 5V pin to provide power, and the GND pin to one of the Arduino's GND pins to complete the circuit. Then, connect the control pin of the servo motor to a PWM-capable digital pin on the Arduino, such as pin 9.

Applications of Servo Motors:

- Robotics: Used for precise control of robot arms and joints.
- Industrial Automation: Controls machinery and conveyor belts for accurate movement.
- RC Vehicles: Manages steering and throttle in remote-controlled cars, boats, and planes.

5.2 SOFTWARE DESCRIPTION:

5.2.1 ARDUINO IDE:

The Arduino integrated development environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in the programming language Java. It is used to write and upload programs to Arduino board.

The source code for the IDE is released under the GNU General Public License, version 2. The Arduino IDE supports the languages [C](#) and C++ using special rules of code structuring.^[4] The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub `main()` into an executable cyclic executive program with the GNU tool chain, also included with the IDE distribution. The Arduino IDE employs the program argued to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware.

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or

a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board

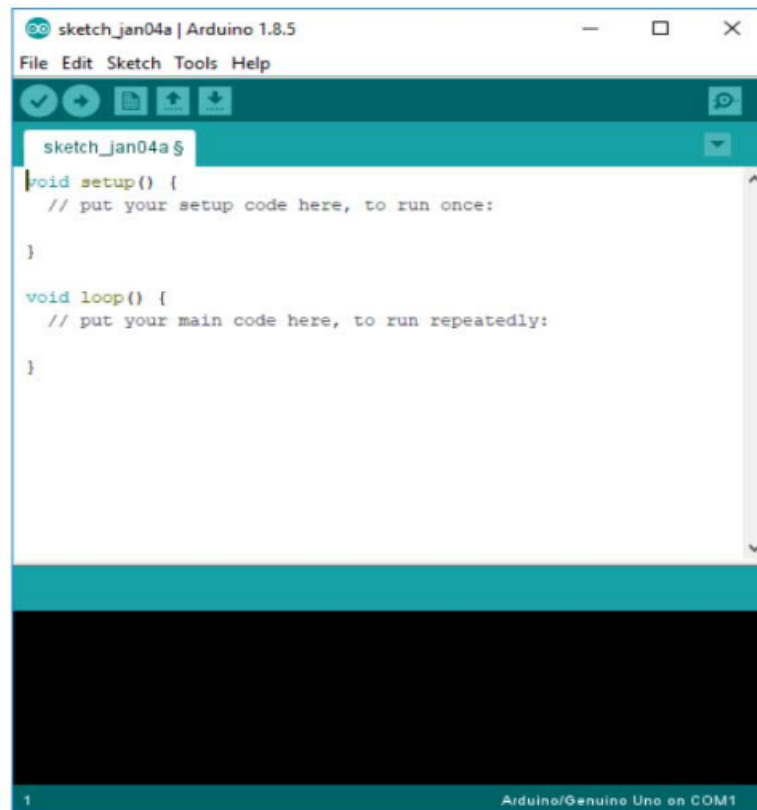


Fig 11: The Arduino IDE

The Arduino IDE is incredibly minimalistic, yet it provides a near-complete environment for most Arduino-based projects. The top menu bar has the standard options, including “File” (new, load save, etc.), “Edit” (font, copy, paste, etc.), “Sketch” (for compiling and programming), “Tools” (useful options for testing projects), and “Help”. The middle section of the IDE is a simple text editor that where you can enter the program code. The bottom section of the IDE is dedicated to an output window that is used to see the status of the compilation, how much memory has been used, any errors that were found in the program, and various other useful messages.

Projects made using the Arduino are called sketches, and such sketches are usually written in a cut-down version of C++ (a number of C++ features are not included). Because programming a microcontroller is somewhat different from programming a computer, there

are a number of device-specific libraries (e.g., changing pin modes, output data on pins, reading analog values, and timers). This sometimes confuses users who think Arduino is programmed in an “Arduino language.” However, the Arduino is, in fact, programmed in C++. It just uses unique libraries for the device.

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuine hardware to upload programs and communicate with them.

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom right hand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

LIBRARIES

Libraries provide extra functionality for use in sketches, e.g. working with hardware or manipulating data. To use a library in a sketch, select it from the Sketch > Import Library menu. This will insert one or more `#include` statements at the top of the sketch and compile the library with your sketch. Because libraries are uploaded to the board with your sketch, they increase the amount of space it takes up. If a sketch no longer needs a library, simply delete its `#include` statements from the top of your code.

There is a list of libraries in the reference. Some libraries are included with the Arduino software. Others can be downloaded from a variety of sources or through the Library Manager. Starting with version 1.0.5 of the IDE, you do can import a library from a zip file and use it in an open sketch.

CONNECTING THE ARDUINO

Connecting an Arduino board to your PC is quite simple. On Windows:

1. Plug in the USB cable - one end to the PC, and one end to the Arduino board.
2. When prompted, select "Browse my computer for driver" and then select the folder to which you extracted your original Arduino IDE download.
3. You may receive an error that the board is not a Microsoft certified device - select "Install anyway."
4. Your board should now be ready for programming.

When programming your Arduino board it is important to know what COM port the Arduino is using on your PC. On Windows, navigate to Start->Devices and Printers, and look for the Arduino. The COM port will be displayed underneath.

Alternatively, the message telling you that the Arduino has been connected successfully in the lower-left hand corner of your screen usually specifies the COM port it is using.

PREPARING THE BOARD

Before loading any code to your Arduino board, you must first open the IDE. Double click the Arduino .exe file that you downloaded earlier. A blank program, or "sketch," should open.

The Blink example is the easiest way to test any Arduino board. Within the Arduino window, it can be found under File->Examples->Basics->Blink.

Before the code can be uploaded to your board, two important steps are required.

1. Select your Arduino from the list under Tools->Board. The standard board used in RBE 1001, 2001, and 2002 is the Arduino Mega 2560, so select the "Arduino Mega 2560 or Mega ADK" option in the dropdown.
2. Select the communication port, or COM port, by going to Tools->Serial Port.

If you noted the COM port your Arduino board is using, it should be listed in the dropdown menu. If not, your board has not finished installing or needs to be reconnected.

LOADING CODE

The upper left of the Arduino window has two buttons: A checkmark to Verify your code, and a right-facing arrow to Upload it. Press the right arrow button to compile and upload the Blink example to your Arduino board.

The black bar at the bottom of the Arduino window is reserved for messages indicating the success or failure of code uploading. A "Completed Successfully" message should appear once the code is done uploading to your board. If an error message appears instead, check that you selected the correct board and COM port in the Tools menu, and check your physical connections.

If uploaded successfully, the LED on your board should blink on/off once every second. Most Arduino boards have an LED prewired to pin 13.

It is very important that you do not use pins 0 or 1 while loading code. It is recommended that you do not use those pins ever.

Arduino code is loaded over a serial port to the controller. Older models use an [FTDI](#) chip which deals with all the USB specifics. Newer models have either a small AVR that mimics the FTDI chip or a built-in USB-to-serial port on the AVR micro-controller itself.

CHAPTER 6

CONCLUSION AND FUTURE WOK

6.1 CONCLUSION

The Hygiene Ensured Smart Door Locking System successfully demonstrates the integration of automation, safety, and hygiene in restroom management. The system achieves touch-free operation by employing infrared sensors, servo-controlled door mechanisms, and automated flushing, thereby minimizing human contact and reducing the transmission of germs. By coordinating door control, flushing, and emergency alert functionalities through an Arduino-based microcontroller, the system ensures smooth and reliable operation while maintaining user privacy and safety. Real-time feedback via an LCD display provides operational transparency, allowing users and maintenance staff to monitor door status, occupancy, flush activity, and emergency alerts effectively.

Through rigorous testing, including unit, integration, functional, performance, and safety evaluations, the system proved to be accurate, reliable, and responsive under various operational conditions. Emergency mechanisms were validated to respond immediately, ensuring user safety in critical situations. The modular design of the system facilitated easy assembly, maintenance, and scalability, allowing installation in multiple restroom environments, including hospitals, public facilities, and smart infrastructure settings.

Economically, the system is feasible due to the affordability of components and reduced long-term maintenance costs, while its energy-efficient design and controlled flushing contribute to sustainability. Socially, the system promotes hygiene awareness, supports accessibility for differently-abled individuals, and aligns with modern expectations for automated infrastructure. Environmentally, water and energy conservation are achieved, complementing sustainable development goals.

The project successfully addresses the challenges of hygiene, user convenience, and operational efficiency in restroom environments. It demonstrates that automation, when integrated thoughtfully, can enhance user safety, reduce contamination risks, and provide a seamless, user-friendly experience. The proposed system establishes a foundation for future

smart restroom solutions, combining technological innovation with practical utility, while being adaptable, scalable, and environmentally responsible. This work validates the concept that automation in hygiene-critical spaces can be implemented efficiently and effectively, benefiting both users and facility operators alike.

6.2 FUTURE WORK

The future development of the Hygiene Ensured Smart Door Locking System focuses on enhancing system intelligence, connectivity, adaptability, and sustainability. One potential advancement involves IoT integration, enabling remote monitoring, data collection, and predictive maintenance. By connecting the system to cloud-based platforms, facility managers can access real-time operational data, track usage patterns, detect malfunctions, and schedule maintenance proactively. Such connectivity would facilitate large-scale deployments in public facilities, airports, and hospitals, optimizing resource allocation and improving user experience.

Another area of future work is advanced sensor integration. While the current system relies on infrared sensors for presence and flush detection, future versions could incorporate ultrasonic sensors, proximity sensors, or vision-based systems to improve accuracy and responsiveness, even under challenging environmental conditions. Gesture recognition and touchless interfaces could further enhance accessibility for differently-abled users or children.

Artificial intelligence and machine learning could be employed to analyze usage patterns and optimize flush timing, energy consumption, and door operations. Predictive analytics could adjust system behavior dynamically based on historical data, peak usage times, or occupancy trends, improving efficiency and reducing operational costs. Additionally, adaptive learning algorithms could detect abnormal usage patterns or emergencies more effectively, providing faster responses.

Sustainability and environmental impact can also be enhanced. Future designs may include smart water recycling systems, sensor-based water flow optimization, and solar-powered operation to reduce energy dependency. Materials for hardware components could be upgraded to eco-friendly, durable alternatives, minimizing waste and maintenance.

Integration with smart building management systems is another promising direction. The restroom system could communicate with lighting, HVAC, or security systems to provide a fully automated, responsive, and energy-efficient environment. Mobile applications could allow users to check restroom availability, status, or cleanliness remotely, improving convenience and overall user satisfaction.

Finally, future work may focus on expanding emergency handling capabilities, including automated notifications to mobile devices, integration with healthcare or security systems, and voice-controlled emergency assistance. Multi-language LCD displays or audio alerts could further enhance accessibility for diverse user groups.

Future developments aim to transform the Hygiene Ensured Smart Door Locking System into an intelligent, connected, adaptive, and sustainable solution for modern restroom management. By leveraging IoT, AI, advanced sensors, and smart infrastructure integration, the system can provide enhanced hygiene, improved user convenience, efficient resource utilization, and proactive safety management. These advancements will make the system more robust, scalable, and aligned with the evolving needs of public, healthcare, and smart facility environments.

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APPENDIX

A.SOURCECODE

```
#include <LiquidCrystal.h>

#include <Servo.h>

// -----

LiquidCrystal lcd(13, 12, 11, 10, 9, 8);

Servo doorServo;

const int irEntryPin = 2;    // First IR: Entry/Exit

const int irFlushPin = 3;    // Second IR: Flush

const int servoPin = 4;      // Servo motor

const int emergencySwitch = 5; // Emergency switch

const int buzzerPin = 7;     // Buzzer pin

// -----

int personCount = 0;         // Counter for IN/OUT

bool flushDone = false;     // Flush status

bool doorOpen = false;

// Timing
```

```
unsigned long doorStartTime = 0;

const unsigned long doorOpenDuration = 2000; // 2 sec

unsigned long flushStartTime = 0;

const unsigned long flushDuration = 1000; // 1 sec


// IR sensor debounce

unsigned long lastIREntryTime = 0;

const unsigned long debounceDelay = 500; // 500ms debounce

unsigned long lastIRFlushTime = 0;


// Status strings

String personStatus = "Person OUT";

String flushStatus = "Flush OFF";

String doorStatus = "DC";

String emergencyStatus = "No Emerg";


// Previous states for edge detection

int prevEntryState = LOW;

int prevFlushState = LOW;


// Flag to track if door was opened after flush
```

```

bool doorOpenedAfterFlush = false;

// -----

void setup() {

  lcd.begin(16, 2);

  doorServo.attach(servoPin);

  doorServo.write(0);    // Door closed initially

  pinMode(irEntryPin, INPUT);

  pinMode(irFlushPin, INPUT);

  pinMode(emergencySwitch, INPUT);

  pinMode(buzzerPin, OUTPUT); // Set buzzer pin as output

  digitalWrite(buzzerPin, LOW); // Ensure buzzer is off initially

  // Display startup message

  lcd.setCursor(0, 0);

  lcd.print(" Smart Toilet ");

  lcd.setCursor(0, 1);

  lcd.print(" Keep Clean ");

  delay(3000); // Allow time for display

  lcd.clear();

  // Read initial sensor states to prevent false triggers

  prevEntryState = digitalRead(irEntryPin);

  prevFlushState = digitalRead(irFlushPin);

```

```

// Additional delay to allow IR sensors to stabilize

delay(1000); // 1-second delay for sensor stabilization
}

// -----

void loop() {

    unsigned long currentTime = millis();

    // Read current states

    int currentEntryState = digitalRead(irEntryPin);

    int currentFlushState = digitalRead(irFlushPin);

    // -----

    // Emergency switch (only active when person is inside)

    if (personCount % 2 == 1 && personStatus == "Person IN") {

        emergencyStatus = digitalRead(emergencySwitch) ? "Emerg" : "No Emerg";

    } else {

        emergencyStatus = "No Emerg"; // Default when no person is inside

    }

    // Control buzzer based on emergency status

    if (emergencyStatus == "Emerg") {

        digitalWrite(buzzerPin, HIGH); // Turn buzzer on

```

```

} else {

    digitalWrite(buzzerPin, LOW); // Turn buzzer off

}

// -----

// ENTRY IR detection (rising edge with debounce)

if (currentEntryState == HIGH && prevEntryState == LOW && currentTime -
lastIREntryTime >= debounceDelay) {

    lastIREntryTime = currentTime;

    if (personCount % 2 == 0) {

        // Person entering

        personCount++;

        flushDone = false;      // Flush required

        personStatus = "Person IN";

        doorServo.write(90);

        doorOpen = true;

        doorStartTime = currentTime;

        doorOpenedAfterFlush = false; // Reset flush door flag

    }

    else if (personCount % 2 == 1 && flushDone) {

        // Person leaving (after flush)

        personCount++;
    }
}

```

```

    personStatus = "Person OUT";

    // Door is already open from flush; reset timer for closing

    doorStartTime = currentTime; // Keep door open for exit

    flushDone = false;      // Reset for next person

    doorOpenedAfterFlush = false; // Reset flush door flag

}

// If flush not done, door stays closed (no action taken)

}

// Update previous state

prevEntryState = currentEntryState;

// -----

// FLUSH IR detection (rising edge with debounce)

if (currentFlushState == HIGH && prevFlushState == LOW && !flushDone &&
currentTime - lastIRFlushTime >= debounceDelay) {

    lastIRFlushTime = currentTime;

    flushDone = true;

    flushStartTime = currentTime;

    flushStatus = "Flush ON";

    // Door does NOT open here; it opens after flush completes

}

// Update previous state

```



```

prevFlushState = currentFlushState;

// -----

// Handle flush duration and open door after flush completes

if (flushDone && currentTime - flushStartTime >= flushDuration) {

    flushStatus = "Flush OFF";

    if (!doorOpenedAfterFlush) { // Open door only once after flush

        doorServo.write(90);

        doorOpen = true;

        doorStartTime = currentTime;

        doorOpenedAfterFlush = true; // Prevent reopening until next cycle

    }

}

// -----

// Close door automatically (only after entry or exit, not after flush)

if (doorOpen && !doorOpenedAfterFlush && currentTime - doorStartTime >=
doorOpenDuration) {

    doorServo.write(0);

    doorOpen = false;

}

doorStatus = doorOpen ? "DO" : "DC";

```

```
// -----  
  
// Update LCD with padding to clear previous text  
  
lcd.setCursor(0, 0);  
  
lcd.print(personStatus + " "); // Pad to cover up to 11 chars  
  
lcd.setCursor(11, 0);  
  
lcd.print(emergencyStatus.substring(0, 5)); // Shorten to fit (e.g., "No Em" or "Emerg")  
  
lcd.setCursor(0, 1);  
  
lcd.print(flushStatus + " "); // Pad to cover up to 11 chars  
  
lcd.setCursor(14, 1);  
  
lcd.print(doorStatus);  
  
}
```

B.SCREENSHOT

